SWOT Tank Water Tracker for Sri Lanka

Oct 14, 2026

Introduction

Sri Lankan farming runs on water stored in tanks and reservoirs, but storage figures are reported mainly for major reservoirs, not small village tanks. In 2018 FAO named the dry zone's cascaded tank-village system (ellanga gammana) a Globally Important Agricultural Heritage System (Daily FT). The Irrigation Department reports storage for its major reservoirs, such as 75% total capacity in December 2023 (Daily FT).

This project builds an AI system that estimates how much water each tank holds, using free satellite data. It combines water area, mapped by deep learning from Sentinel-1 and Sentinel-2, with water height from SWOT, a NASA and French space agency (CNES) satellite launched in December 2022.

Farmers and irrigation officials can then see storage for every tank in a cascade and plan the next Yala or Maha crop.

Problem Statement

Satellites already show where tank water is, but not how much water each tank holds.

- **Area is not volume.** IFAD's 2026 maps track the water surface of more than 10,650 water bodies in six districts at 10 m resolution, but not their volume (IFAD). A shallow tank and a deep tank of the same area hold very different amounts of water.
- **Water height used to be out of reach.** Earlier altimeter satellites measured only along the line under their path; SWOT's wide swath covers far more ground (AVISO) and is specified for lakes down to 250 m × 250 m (NASA ARSET).
- **Clouds and water plants hide the water.** Optical images fail under monsoon clouds, and floating plants cover water surfaces; in North Bolgoda Lake water hyacinth spread most from May to November (Dawalagala et al., 2023).
- **No tool joins these for Sri Lanka.** No study found combines SWOT water heights with AI water maps to give storage for Sri Lankan tanks.

Research Questions and Objectives

The project tests whether free satellite data and AI can report water storage for every tank in a cascade, not just the major reservoirs.

Research questions

1. How accurately can deep learning map tank water from Sentinel-1 and Sentinel-2, including tanks under cloud or partly covered by water plants?
2. How accurately does SWOT measure water height on Sri Lankan tanks of different sizes, compared with gauge readings?
3. Does combining AI water area with SWOT height give storage close to official and measured values?
4. Can storage be forecast 2–4 weeks ahead from recent storage and rainfall? (stretch)

Main objective: build and validate a system that reports storage for every tank in one dry-zone cascade and for 2–3 major reservoirs, using only free satellite data.

Specific objectives

- Build a labelled dataset of Sri Lankan tank water masks, including cloudy and plant-covered cases.
- Train a deep-learning model that maps water through clouds and under floating plants.
- Process SWOT water heights for the study tanks and check them against gauge readings.
- Build area–height–volume curves per tank and estimate storage over time.
- Deliver a dashboard showing storage per tank and per cascade.

Literature Review and Research Gap

SWOT storage has been shown on large US reservoirs and, combined with other satellites' water area, on 245 reservoirs in five countries; SWOT heights have been checked on Taiwanese ponds, while Sri Lankan work maps water area only. We found no study combining them for Sri Lankan tanks.

Study	Where	What it does	Gap for this project
Ma et al., 2026	12 reservoirs, western USA	Uses SWOT height and area together to estimate storage; height error under 20 cm, storage error under 10%	Large US reservoirs; not small tanks, not Sri Lanka

Study	Where	What it does	Gap for this project
Das and Hossain, 2025	245 reservoirs in the USA, Spain, India, South Africa and Thailand	Joins SWOT storage with Sentinel-1, Sentinel-2 and Landsat water area, using a Random Forest to calibrate the non-SWOT data, to track storage change	Not deep learning; not Sri Lanka
Wang and Tseng, 2025	14 reservoirs and 15 ponds, Taiwan	Checks SWOT heights against gauges; a refined workflow cut errors to 0.15–0.53 m and detected ponds of about 110 × 110 m	Heights only; no volume, no AI
IFAD and ESA, 2026	Six districts, Sri Lanka	Monthly 10 m water-surface maps of more than 10,650 tanks, 2017 to August 2025	Area only; no volume
Wu et al., 2023	Four small reservoirs, California	A ResNet34 model reads water level from Sentinel-1 and Sentinel-2 images; error 0.19–0.77 m	Needs gauge records to train for every reservoir; no SWOT
Madhurshan et al., 2024 (UoM)	Victoria reservoir	Compares water area from Landsat 8, Sentinel-1 and Sentinel-2	One reservoir; area only
Meyen, 2018 (UoM MSc)	Kotmale reservoir	Forecasts water level with LSTM and GRU from gauge history and weather data	Needs gauges; no satellite
Jayawardena et al., 2021 (UoM)	Sri Lanka	Extracts water from Landsat 8 with classic classifiers; kappa above 0.95	Classic methods on Landsat; area only

This project's contribution: the first published SWOT storage study for Sri Lankan tanks, checked against local gauge readings, plus an AI water map that handles clouds and water plants and updates storage every 5–12 days. SWOT's own lake product already reports storage change for listed lakes; this project adds the AI maps, small tanks and ground checks.

Study Area and Data Sources

The study covers one dry-zone tank cascade plus 2–3 major reservoirs with published storage, chosen only after checking which tanks SWOT actually observes.

Choosing the study area

- Tanks inside SWOT's swath, seen on at least two passes per 21-day cycle.
- Tanks listed in SWOT's prior lake database, which includes water bodies down to 10,000 m² (NASA ARSET); several larger than 250 m × 250 m, plus smaller ones to test the limits.
- Gauge posts the team can reach, and a cooperative farmer organisation or irrigation office.
- Anuradhapura District is a strong candidate: the Palugaswewa area alone has 12 cascades (Daily FT), and the Irrigation Department reports storage for Hurulu Oya reservoir (Daily FT).

Data

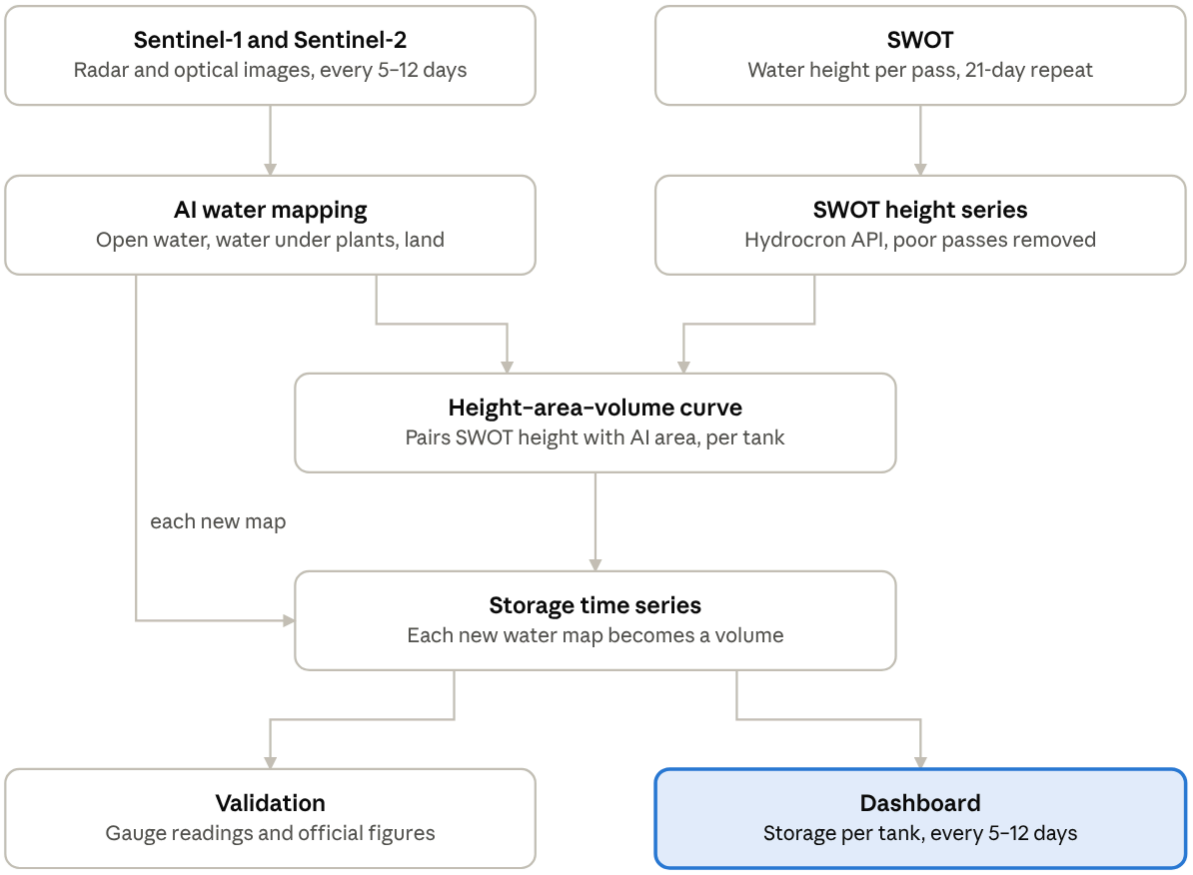
Data	Source	What it gives	Use in the project
SWOT lake product (LakeSP)	NASA PO.DAAC; the Hydrocron API returns time series by lake ID (PO.DAAC webinar)	Water height and area per pass, 21-day repeat	Water height for volume
SWOT pixel cloud (PIXC)	NASA PO.DAAC	Raw water pixels with heights	Smaller tanks missing from the lake product (stretch)
Sentinel-1 radar	Copernicus, via Google Earth Engine	Images that see through cloud	Water mapping in the monsoon
Sentinel-2 optical	Copernicus, via Google Earth Engine	10 m colour and infrared images	Water and water-plant mapping
Rainfall	CHIRPS or GPM IMERG	Daily rainfall	Storage forecasting (stretch)

Data	Source	What it gives	Use in the project
Official storage	Irrigation Department reports	Storage of major reservoirs	Validation
Gauge readings	The team, at 5–10 tanks on SWOT pass dates	Water level in the field	Validation
Historical gauge records	Irrigation Department, 2023 to 2026	Daily water levels of major and medium reservoirs	Checking SWOT across its whole record

Methodology

SWOT passes build each tank's height–area curve; after that, every new Sentinel water map converts straight into storage, so updates come every 5–12 days instead of every 21.

SWOT builds each tank's curve; Sentinel maps then update storage every 5–12 days



method pipeline · two satellite inputs, one storage output

The left path runs every 5–12 days; the SWOT path only needs enough passes, from its record since July 2023, to fit each tank's curve.

1. **Tank inventory:** list every tank in the cascade with its outline and SWOT lake ID, and record which SWOT passes cover it.
2. **Labelled dataset:** create water masks for many dates per tank. Start from water indices on clear Sentinel-2 images, correct them by hand, and mark three classes: open water, water under plants, and land.
3. **AI water mapping:** train a segmentation network (for example U-Net) on Sentinel-1 and Sentinel-2 together, so it still works when clouds block Sentinel-2. The tank's recent history helps it spot water hidden under plants.
4. **SWOT heights:** pull each tank's full height record since July 2023 through the Hydrocron API and drop low-quality passes. Tanks without a Prior Lake Database ID need the SWOT pixel cloud instead.
5. **Height–area–volume curves:** pair each SWOT height with the AI water area from the same date, fit a curve per tank, and integrate it to get volume. Ma et al., 2026 used SWOT's simultaneous height and area the same way. SWOT only sees levels a tank has reached since 2023, so the curve gives storage above the lowest level seen; Irrigation Department capacity tables, or a fitted bowl shape, give total storage and % full.
6. **Storage time series:** convert every new water map into height and volume through the curve, and report storage change, plus % of full capacity where the capacity is known.
7. **Forecasting (stretch):** predict storage 2–4 weeks ahead from recent storage and rainfall.
8. **Dashboard:** a map of the cascade with each tank's current storage, trend and last update.

Evaluation Plan

Every result is checked against field or official data, and every AI model is compared with a simpler baseline.

What is tested	Metric	Compared against
Water mapping	IoU and F1 for water; reported separately for cloudy and plant-covered tanks	Water-index thresholds on Sentinel-2, a Sentinel-1 threshold method, and IFAD/ESA maps where shared
SWOT water height	Error in cm against gauge readings on pass dates, by tank size	SWOT specification: 10 cm for water bodies over 1 km², 25 cm between 250 m × 250 m and 1 km² (NASA ARSET)

What is tested	Metric	Compared against
Storage	Error as a percentage of capacity against Irrigation Department figures and gauge-based volumes	Area-only estimate that assumes a fixed depth
Update frequency	Days between storage updates	SWOT alone (21-day repeat)
Forecast (stretch)	Error at 2 and 4 weeks ahead	"No change" baseline
Usefulness	Feedback from irrigation officers and farmer organisations	—

Expected Outcomes and Contributions

The project delivers the first published, gauge-checked SWOT storage figures for Sri Lankan tanks, plus the data and tools to repeat them.

1. **Evidence:** SWOT's accuracy on Sri Lankan tanks, measured against gauges and broken down by tank size.
2. **Model and dataset:** an AI water map that works under clouds and floating plants, with the labelled Sri Lankan tank dataset behind it.
3. **Storage curves:** a height–area–volume curve for every studied tank, reusable by irrigation engineers.
4. **Tool:** a dashboard showing storage for every tank in one cascade, updated every 5–12 days.
5. **Publication:** a paper for a conference such as UoM's ICITR.

Scope

The core is deliberately limited to one cascade, because the full idea is larger than a typical final-year project. UoM IT final-year computer vision papers usually fine-tune a pretrained model on about 600–1,250 images and add a web app (ICITR 2024), while most UoM satellite studies use one satellite and classic classifiers. This project sits above the first and close to Master's-level satellite work.

Level	Included	Why
Core (must deliver)	One cascade plus 2–3 major reservoirs; AI water mapping with a water-plant class; SWOT heights; height–area–volume curves; storage time series; validation; simple dashboard	Answers research questions 1–3 end to end
Stretch (if time allows)	Storage forecast 2–4 weeks ahead; SWOT pixel cloud for tanks missing from the lake product; cloud-removal module; a second cascade	Adds research question 4 and wider coverage
Out of scope	Whole-island coverage; mobile app; real-time alerts; surveys of tank beds	Too large for 3 people in 8 months

Tools, Equipment and Data Access

All data and software are free; the only real cost is travel to the study area for gauge readings.

Type	Item	Purpose
Data access	NASA Earthdata account	SWOT lake data through PO.DAAC and the Hydrocron API
Data access	Google Earth Engine	Sentinel-1, Sentinel-2 and rainfall data
Software	Python, PyTorch, xarray, rasterio, geopandas	Models and satellite data processing
Software	QGIS	Checking maps and drawing tank outlines
Software	FastAPI, Next.js, Leaflet	Dashboard
Compute	Google Colab GPU and team laptops	Training and testing
Field	Team smartphones (camera and GPS)	Photographing gauge posts on SWOT pass dates

Team Roles

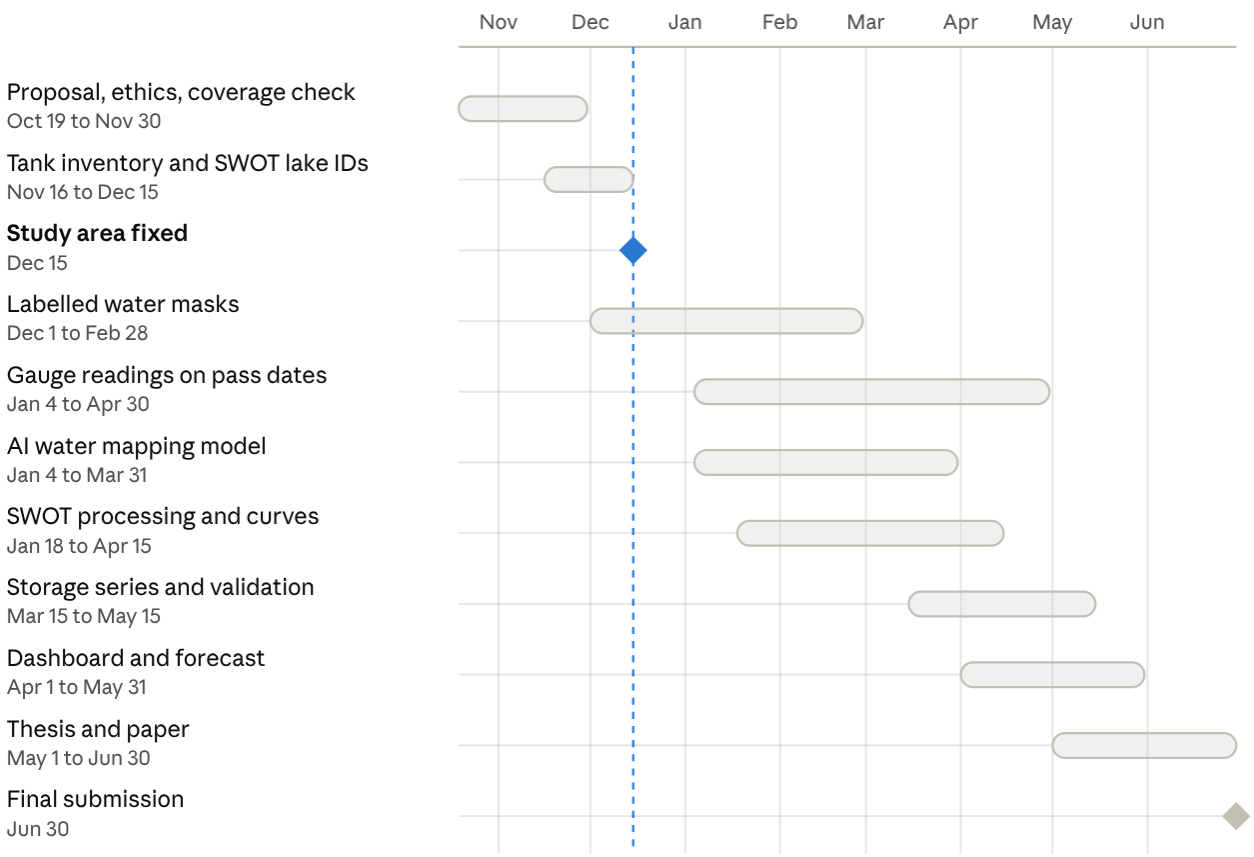
Each member owns one research question end to end; all three share field trips, labelling and writing.

Member	Owns	Key outputs	Research question
Member A: water mapping	Labelled dataset, Sentinel-1 and Sentinel-2 segmentation model, water-plant class	Water maps for every tank and date	1
Member B: SWOT and storage	SWOT processing, gauge validation, height–area–volume curves	Storage time series and accuracy report	2 and 3
Member C: forecasting and dashboard	Data pipeline, storage forecast, dashboard, user feedback	Working dashboard and forecast results	4, plus whole-system testing

Timeline

The study area is fixed by 15 December, once SWOT coverage is confirmed

Proposed plan, October 2026 to June 2027; dates assumed, adjust to faculty deadlines



project timeline · 9 phases, 2 milestones

Gauge readings run from January to April while the models are built, so validation data is ready when storage estimates arrive in March. Historical gauge records since 2023 are requested in month one, because January to April alone gives only a few SWOT passes.

Risks, Ethics and Safety

The biggest risk is SWOT not seeing the chosen tanks often enough, so coverage is checked before the study area is fixed.

Risk	Mitigation
SWOT passes rarely, or not at all, over the chosen tanks	Check pass coverage and lake IDs first; pick the cascade with the most passes

Risk	Mitigation
Small tanks missing from SWOT's lake product	Count Prior Lake Database IDs before choosing the cascade; if many small tanks lack one, move the pixel cloud from stretch to core
No access to official gauge records	Request them through the supervisor in the first month; photograph gauge posts on pass dates
Floating plants hide water	Separate water-plant class plus the tank's history; report accuracy for these tanks separately; Sentinel-1 alone can mistake dense plant cover for land, so clear Sentinel-2 dates anchor these tanks
Monsoon clouds	Sentinel-1 radar as an input; Sentinel-2 used when clear
Field safety	Visit in daylight and in pairs, read gauges from the bund, never enter the water; dry-zone tanks can have elephants and crocodiles nearby
Permission	Inform the irrigation office and farmer organisation before any visit
Data volume	Process in Google Earth Engine and through the Hydrocron API; download only the study area

References

Sri Lankan context

- Daily FT (2018). Sri Lanka designated as a globally important agricultural heritage site.
- Daily FT (2023). Total water capacity of irrigation reservoirs estimated at 75%.
- IFAD (2026). Water surface maps to manage ancient tank cascade systems in Sri Lanka.

University of Moratuwa studies

- Madhurshan, R., Dhananjaya, K. I. K., Keerthi, T., Dassanayake, S. M., Chao, Z. (2024). Multi sensor satellite remote sensing data based comparative assessment of the surface water level dynamics of the Victoria reservoir, Sri Lanka.
- Meyen, A. N. (2018). Predicting reservoir water levels using deep learning. MSc thesis.

- Jayawardena, C. L., Brinthan, K., Thanujan, T. (2021). Evaluation of machine learning algorithms in classifying multispectral imagery on waterbody extraction.
- Dawalagala, H. S., Radampola, A., Gowsigan, P. T., Chaminda, S. P., Dassanayake, S. M., Jayawardena, C. L. (2023). Comparative analysis of water hyacinth dynamics in North Bolgoda Lake, Sri Lanka.
- University of Moratuwa (2024). ICITR 2024 abstracts.

SWOT and related methods

- Ma, X., Wang, J., Gharari, S., Mizukami, N., Lettenmaier, D. P. (2026). Monitoring reservoir storage using SWOT satellite observations and a reservoir operation model. *Water Resources Research*, 62(3).
- Wang, C.-H., Tseng, K.-H. (2025). Potential of SWOT mission in monitoring Taiwan surface water dynamics. ACRS 2025.
- Wu, J., et al. (2023). Combining satellite imagery and a deep learning algorithm to retrieve the water levels of small reservoirs. *Remote Sensing*, 15(24), 5740.
- NASA ARSET (2025). SWOT training, Part 1 Q&A.
- NASA PO.DAAC (2025). SWOT hydrology tools webinar.
- AVISO. SWOT science orbit.
- Das, P., Hossain, F. (2025). Multi-satellite tracking of surface water storage change in the era of the Surface Water and Ocean Topography (SWOT) satellite mission. *Earth and Space Science*, 12, e2024EA004178.

Research notes and extra sources: Research notes

Research notes

Checked Oct 14, 2026. Three findings matter most: SWOT data degrades in heavy rain, ready-made SWOT time series exist only for tanks in its prior lake list, and a pretrained water model and official tank outlines already exist to start from.

Key facts to quote

Fact	Figure	Source
Small tanks in Sri Lanka	About 23,000 tanks of 80 ha or less in the National Tanks Survey; about 21% non-functional	Lopane et al., 2025
Tanks with satellite water maps	More than 10,650 water bodies, monthly at 10 m, 2017 to August 2025	IFAD, 2026
Worst water plants in seasonal tanks (Anuradhapura District)	Water hyacinth, lotus and Salvinia	Munasinghe and Sundarabarathy (IWMI)
Silt	Sediment makes tanks shallower and cuts storage	Munasinghe and Sundarabarathy (IWMI)
SWOT accuracy targets	Water height within 10 cm for lakes over 1 km² and 25 cm for lakes from 250 m × 250 m to 1 km² (open water); area within 15%	SWOT Science Team, 2024
Published reservoir levels	The Mahaweli Authority reports levels against spill, for example Victoria 188 cm below spill on 8 January 2026	Ada Derana

SWOT limits and how to handle them

SWOT is not a perfect gauge: rain, calm water and hills can spoil a pass, so the project keeps only clean passes and leans on Sentinel-1 in the monsoon.

Limit	What happens	How the project handles it
Heavy rain	SWOT's Ka-band radar is more sensitive to rain than the Ku-band radars on most earlier altimeters, and heavy rain degrades the data; the ocean product has a rain flag, the lake product has none (AVISO)	Use quality flags and skip passes on heavy-rain days (from rainfall data); fit curves from dry-season passes; use Sentinel-1 for monsoon water area
Dark water	Calm, sheltered water reflects weakly; a known processing issue with a fix planned (SWOT Science Team, 2024)	Use quality flags; report affected tanks separately
Layover in hills	Nearly all land is in layover, though often with little impact (SWOT Science Team, 2024)	Prefer flat dry-zone cascades; treat hill reservoirs such as Victoria as a separate test
Specular ringing	Very bright reflections smear into nearby pixels	Use quality flags
Only listed lakes in Hydrocron	Ready-made time series exist only for lakes in SWOT's prior lake database (NASA Earthdata)	Check each tank's lake ID first; use the pixel cloud for tanks that are missing (stretch)

Ready-made starting points

The team does not start from zero: a pretrained water model, labelled tiles, a global storage dataset and official tank outlines already exist.

Resource	What it gives	Use in the project
Prithvi-EO-2.0 Sen1Floods11 model	IBM and NASA water and flood segmentation model using six Sentinel-2 bands; Apache-2.0 licence	Starting model and baseline for the AI water map
Sen1Floods11	446 hand-labelled 512 × 512 tiles at 10 m, Sentinel-1 and Sentinel-2	Pre-training

Resource	What it gives	Use in the project
GloLakes	Storage time series for more than 170,000 lakes (1984–2020), with near-real-time monitoring of 27,000; lakes over 1 km² only; volume from a terrain-slope model, not bathymetry	Comparison for the major reservoirs
UNDP tank outlines	Satellite-traced tank outlines used by Lopane et al., 2025	Tank inventory; request from UNDP
Hydrocron	SWOT time series as CSV or GeoJSON, by lake ID	SWOT water heights

Partners and data to request

Ask through the supervisor in the first month, because official records are the slowest part to get.

Who	Ask for
Irrigation Department	Daily water levels and capacity tables for major and medium reservoirs in the study area
Mahaweli Authority	Reservoir levels against spill for the Mahaweli reservoirs
Department of Agrarian Development	Minor tank lists, gauge posts and farmer organisation contacts
UNDP Sri Lanka	The satellite-traced tank outlines
IWMI (based in Colombo)	Advice on tank cascades; possible co-supervision

Checks before the proposal

- [ ] Query Hydrocron for the candidate tanks' lake IDs and pass dates
- [ ] Count SWOT passes per 21-day cycle over Anuradhapura District
- [ ] Check which Sri Lankan reservoirs appear in GloLakes
- [ ] Request reservoir level records through the supervisor
- [ ] Run the Prithvi water model on a few Sri Lankan tanks to see where it fails

Sources

- Lopane et al. (2025). Optimising water storage for climate resilience: geospatial targeting for small tanks rejuvenation in Sri Lanka. *Journal of Environmental Management*, 374.
- IFAD (2026). Water surface maps to manage ancient tank cascade systems in Sri Lanka.
- Munasinghe, D., Sundarabarathy, T. V. Utilization of aquatic plants. IWMI.
- SWOT Science Team (2024). HR requirements status and plan.
- AVISO. Winds, waves and rain seen by SWOT and nadir altimeters.
- NASA Earthdata (2024). Hydrocron time series API for SWOT hydrology data.
- IBM and NASA. Prithvi-EO-2.0-300M-TL-Sen1Floods11.
- Cloud to Street. Sen1Floods11.
- Hou, J., et al. (2024). GloLakes: water storage dynamics for 27 000 lakes globally from 1984 to present. *Earth System Science Data*, 16.
- Ada Derana (2026). Mahaweli Authority issues update on water levels of major reservoirs.